EX-02-Cross-Platform-Prompting-Evaluating-Diverse-Techniques-in-AI-Powered-Text-Summarization

"The Basics of Blockchain Technology"

1. Introduction

Blockchain Technology is a modern technology used to record and store digital information securely. It works as a distributed digital ledger that keeps records of transactions across many computers. Because the data is shared among many systems, it becomes difficult for anyone to change or manipulate the information.

2. How Blockchain Works

Blockchain stores information in small units called blocks. Each block contains transaction data, a timestamp, and a unique identification code known as a hash. These blocks are connected in a sequence, forming a chain. When a new transaction happens, it is verified and added as a new block to the chain.

3. Security in Blockchain

One of the most important features of blockchain is security. It uses **Cryptography** to protect the data stored in each block. This method ensures that information cannot be easily changed or hacked, making blockchain a reliable system for storing sensitive data.

4. Decentralization and Transparency

Blockchain is decentralized, meaning there is no single authority controlling the system. Instead, the network participants share control and verify transactions together. This process increases transparency because every participant can view the transaction history stored on the blockchain.

5. Applications of Blockchain

Block chain technology is widely known for its use in digital currencies like **Bitcoin**. However, it is also used in many other fields such as banking, healthcare, supply chain management, and digital identity verification. These applications help improve efficiency and security in data management.

6. Conclusion

In conclusion, blockchain technology is an innovative system that provides security, transparency, and trust in digital transactions. As technology continues to develop, blockchain is expected to play an important role in many industries and improve the way information and transactions are managed in the digital world.

CHAT GPT:

Zero-shot Prompt :

“Explain the basics of blockchain technology in simple terms, including its definition, how it works using blocks and hashes, the role of cryptography in security, decentralization and transparency, common applications such as Bitcoin and other industries, and its overall importance in improving secure digital transactions.”

Few-shot Prompt:

“The explanation about Blockchain Technology describes it as a secure distributed ledger that records transactions across many computers. Data is stored in blocks linked together with hashes to form a chain. The system uses Cryptography to protect information and ensure it cannot be easily altered. Blockchain is decentralized and transparent because participants verify transactions together. It is commonly used in digital currencies such as Bitcoin, as well as in banking, healthcare, and supply chain management.”

Chain-of-thought:

Blockchain technology is a secure and decentralized digital ledger that records transactions across many computers. It stores data in connected blocks protected by cryptography, making the information transparent and difficult to change. Blockchain is widely used in Bitcoin and other applications such as banking, healthcare, and supply chain management to improve security and efficiency.

Role-based:

Blockchain technology is a secure digital ledger that records transactions across many computers. Information is stored in blocks connected in a chain and protected using cryptography. Because the system is decentralized, no single authority controls it and all participants can view transactions, which increases transparency. Blockchain is widely used in Bitcoin and other industries like banking, healthcare, and supply chain management to improve security and efficiency in managing digital data.

Gemini:

Zero-shot Prompt:

"Act as a technology expert. Summarize the fundamental principles of Blockchain technology. Your explanation must cover its structure (blocks and hashes), the role of decentralization in security, and its practical applications beyond cryptocurrency. Ensure the tone is professional and the information is presented in a clear, bulleted list for a non-technical audience."

Few-shot Prompt:

Blockchain technology is a decentralized digital ledger designed to record transactions across a global network of computers. Unlike traditional databases managed by a single authority, it stores data in "blocks" that are secured through cryptography and linked together to form an immutable chain.

Chain-of-thought:

 First, identify the core definition and the physical structure of a block.

 Second, explain the mechanism that connects these blocks and ensures their security.

 Third, determine the organizational difference between this system and a traditional centralized one.

 Finally, synthesize these points into a concise summary that explains why this technology is impactful across different industries.

Role-based:

"Act as a Chief Technology Officer (CTO) presenting to a board of non-technical investors. Summarize the provided text to explain why our company should trust Blockchain for data integrity. Focus on the structural security of 'blocks,' the shift from centralized to decentralized control, and the diverse industrial applications beyond just Bitcoin. Your goal is to build confidence in the technology’s reliability and transparency."

Deep-seek:

Zero-shot Prompt:

Blockchain is a secure, decentralized digital ledger that records transactions across a network of computers. It uses cryptography and a chain of connected data blocks to ensure transparency, prevent tampering, and enable applications beyond cryptocurrency, such as in banking and healthcare.

Few-shot Prompt:

Blockchain is a secure,decentralized digital ledger that records transactions across a network of computers. It works by linking data in cryptographically secured blocks, which prevents tampering and ensures transparency. This technology is foundational for cryptocurrencies like Bitcoin and has growing applications in sectors such as banking, healthcare, and supply chain management.

Chain-of-thought:

Blockchain is a secure,decentralized digital ledger that records transactions across a network of computers. It links data in cryptographically protected blocks, forming an unchangeable chain that is verified by network participants rather than a single authority. This structure ensures security and transparency, enabling applications beyond cryptocurrencies like Bitcoin to transform industries such as banking and healthcare.

Role-based:

Blockchain is a shared digital ledger that records information across a network, making data both secure and tamper-proof. It works by grouping transactions into linked, cryptographically protected blocks that cannot be altered once verified by the network's participants. Because no single entity controls it, the system offers high transparency and trust. While it is the foundation for cryptocurrencies like Bitcoin, its real business value lies in improving efficiency and security for applications like banking, healthcare, and supply chain management.

platform goals	CHATGPT	GEMINI	DEEP SEEK
ACCURANCY	5	4	4
COHERENCE	4	5	4
SIMPLICITY	4	4	5
SPEED	4	5	5
USER EXPERIENCE	5	5	5